Python Programming

Please follow below rules while designing automation script as

- Accept input through command line or through file.
- Display any message in log file instead of console.
- For separate task define separate function.
- For robustness handle every expected exception.
- Perform validations before taking any action.
- Create user defined modules to store the functionality.

Please add below features in our project named as Platform Surveillance System

1. Add Thread Monitoring Feature

For each running process, display:

- Process Name
- PID
- Number of Threads created by that process

Requirement

Store information in log file along with timestamp.

2. Add Open Files Monitoring Feature

For each process, display:

- Number of files opened by the process

Requirement

- Count open file descriptors using system/library calls
- Handle permission errors properly
- Mention "Access Denied" in log if required

3. Add Actual Memory Allocation Feature

Display real memory usage of each process:

- RSS (Resident Set Size – actual RAM used)
- VMS (Virtual Memory)
- Memory Percentage

Requirement

Show:

- Top 10 memory consuming processes

4. Add Periodic Email Reporting Feature

Automatically send system report through email at regular intervals.

Email must contain:

- Log file attachment
- Summary of:
  - Total processes
  - Top CPU usage processes
  - Top Memory usage processes
  - Top Thread count processes
  - Top Open file processes

Usage

```
PlatformSurveillance.py "MarvellousLogs" "receiver@gmail.com" 10
```

Where:

- MarvellousLogs → log folder
- receiver@gmail.com → receiver mail
- 10 → interval in minutes

Expected Output in Log File

Each process entry should include:

```
Process Name
PID
CPU %
Memory (RSS)
Threads Count
Open Files Count
Timestamp
```